CT-361 Artificial Intelligence & Expert System

Complex Computing Activity (CCA) through Problem Based Learning (PBL)

Course Learning Outcome:
CLO 3 (C3): Identify appropiate AI techniques to solve classification problems of moderate complexity.

Complex Computing Activity Attributes:

- **CA 2: Problem Analysis and Solution Development:** Students analyze the given AI problem, identify relevant inputs and outputs, evaluate alternative solution strategies, and develop an effective computational solution.
- **CA 3: Application of AI Knowledge:** Students demonstrate a thorough understanding of Artificial Intelligence concepts and apply appropriate algorithms and computing principles to implement accurate solutions and explain their practical applications.

Project Title & Problem Statement:

Students must identify a real-world problem in areas such as healthcare, cybersecurity, finance, environmental monitoring, marketing, or smart systems that can be solved using AI or Expert System techniques. The project title should be clear and relevant. The problem statement should briefly describe:

- The problem context and objectives
- Its real-world significance
- The proposed AI or Expert System approach (e.g., rule-based systems, predictive modeling, NLP, object detection, AI agents, chatbots, or hybrid AI approaches.

Problem Description:

The project should address a meaningful real-world problem where intelligent decision-making, prediction, or automated reasoning is required. Students must clearly describe the selected domain, the specific use case, and the goals of the system. The solution should demonstrate how AI or Expert System techniques can transform raw data or domain knowledge into accurate predictions, recommendations, or automated decisions.

At least two datasets/one large, or knowledge sources must be used and briefly described (source, type of data, and relevance). Students should explain how these will be utilized to build and evaluate the proposed AI system using suitable techniques and performance metrics.

Assumptions & Constraints:

- Projects must be completed in designated groups.
- Use at least two relevant datasets or one sufficiently large and suitable dataset for the project.
- Datasets may contain noise, missing values, class imbalnced or inconsistencies; appropriate pre-processing is required.
- Systems should be able to handle diverse data types depending on the chosen domain.
- External factors (e.g., environment, user behavior, dynamic input) may influence system performance.
- All models must adhere to data privacy regulations (e.g., anonymizing personal/sensitive data).
- Ethical considerations must be followed—ensure fairness, transparency, and accountability.
- A baseline model must be implemented, with performance improvement using multiple AI approaches.
- The performance of real-time monitoring depends on network availability

Deliverables:

- **Python Code Implementation:** ***Design*** a complete AI solution for a real-world problem using datasets, including preprocessing, model building, evaluation, and performance improvement (e.g., tuning or ensembling).
- **Visualization:** ***Develop*** a simple, user-friendly visualization or interface to present results. This could include charts, maps or prediction outputs. The goal is to make the findings interpretable and actionable for users.

NED University of Engineering and Technology

Department of Computer Science and Information Technology

Bachelor of Science in Computer Science (Specialization in Artificial Intelligence)

Course Code and Title: CT 361 (Artificial Intelligence & Expert System)

Student Name: _____Roll Number: _____

Cognitive Domain Assessment Rubric – Level C3 (For Final Lab / CCA Evaluation)			
Criterion	**Level of Attainment**		
	0	**1**	**2**
Problem Identification	Not able to correctly identify the problem(s) and required data (input and output)	Able to Partially identify the problem(s) and the required data (input and output)	Able to correctly identify the problem(s) and all the required data (input and output)
Problem Solving Skills	Uses an Ineffective strategy to solve the problem(s)	Use an adequate strategy to solve the problem(s)	Use an effective strategy to solve the problem(s)
Implementation	Incomplete Implementation	Adequate Implementation	Outstanding Implementation
Deliverable(s)	Cannot generate the desired output	Generates partially correct and partially complete output	Generates correct and complete output
Depth of Knowledge	Does not grasp the concept covered and cannot identify real-life application	Fair grasp of concepts covered and can vaguely identify a real-life application	Thorough grasp of the concepts covered and can apply it in real-life situations

Weighted CLO (Cognitive) Score	
Remarks	
Instructor's Signature with Date	